

Jiajun Xia

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Education

University of California, Berkeley

Aug 2026 – May 2027

3+1+1 Joint Program, Civil & Environmental Engineering

Tongji University

Sep 2022 – Jul 2027

B.E. in Transportation Engineering | GPA: 4.53/5.00 | Ranked among top 2 students in the program

Research & Projects

Intelligent Train Driving Assistance under Variable Speed Restrictions

Mar 2026 – Jul 2026

- Developed a real-time driving-assistance framework using DP-generated expert policies, LightGBM behavior cloning, ATP safety validation, and SHAP explanations.
- Achieved **13 ms** inference, **98.43%** classification accuracy, and **>1,000×** faster computation than online dynamic programming.

Data-Driven Metro Travel-Time Reliability Assessment

Oct 2024 – Jun 2025

- Analyzed one month of Shanghai Metro AFC data; developed CoV, TTI, BTI, and PTI and applied Isolation Forest and regression to assess OD-level reliability.
- Achieved pairwise metric correlations **>0.80** and regression $R^2 = 0.742$ (MSE = 0.00287), identifying free-flow travel time and passenger volume as key factors.

Scenario-Adaptive Multi-Vehicle Cooperative Perception

Nov 2022 – Apr 2024

- Collected and analyzed data from **30 CARLA–SUMO–OpenCDA simulation configurations** across five urban and highway scenarios.
- Evaluated **13 BEV cooperative perception models** with uncertainty analysis and NAS-based tuning; reduced perception loss from 29.29 to 0.00568 and improved accuracy across IoU thresholds.

Experience & Training

ZTO Express Transportation Management Intern

Jun 2026 – Aug 2026

Shanghai

- Developed an AI workflow and standardized HTML template for automated transit-time performance reporting.
- Implemented data validation and self-check mechanisms, reducing report generation from **30–60 minutes to under 3 minutes** across five pilot regions.

AWS Educate AI & Machine Learning Program

Jan 2025 – Apr 2025

Remote

- Completed Machine Learning Foundations training and developed an AI agent on the AWS platform.

T-IES Summer School Summer School Participant

Jul 2024 – Sep 2024

Tübingen, Germany

- Completed the program and received the **Outstanding Student Certificate**.

Skills

Programming: Python, MATLAB, SQL, R, C++, $\LaTeX$ | **Simulation & Modeling:** CARLA, SUMO, AutoCAD

Engineering Tools: SolidWorks, Multisim, PTC Creo, Arduino

Honors & Awards

2026	Chinese Government Scholarship
2025	Scholarship, Sichuan Jingchuan Highway Engineering (Group) Co., Ltd.
2025	Chinese Government Scholarship (Endeavor Scholarship)
2025	Silver Award, International College Student Innovation Competition Final
2024	Tongji University Scholarship (Third Prize)
2024	Third Prize, 23rd TongLuRen Transportation Technology Competition